

IMSE 586: Big Data Analytics and Visualization
Fall 2025
Project Progress Report

**Crime Pattern Analysis and Predictive Policing:
A Data-Driven Approach to Public Safety in Detroit**

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Completed Work

Since the submission of our proposal, the team has made substantial progress toward addressing two of the four research questions. We have (i) completed data acquisition, cleaning, and feature engineering, (ii) performed temporal crime pattern analysis as Research Question 1(RQ1), and (iii) conducted spatial hotspot analysis as Research Question 2(RQ2). [1, 2, 3, 4].

Data Acquisition, Cleaning, and Feature Engineering

We collected two primary datasets from the [Detroit Open Data Portal](#):

- **RMS Crime Incidents:** incident-level records with timestamps, spatial coordinates, crime types, and precinct identifiers.
- **Liquor Licenses:** locations of alcohol-serving establishments, used to study crime–alcohol spatial relationships [5].

Data cleaning involved removing duplicate records, dropping entries missing critical temporal or spatial attributes, and standardizing timestamp fields via `pandas.to_datetime()` method.

We introduced several feature groups to support both descriptive and predictive analysis:

- **Temporal features:** year, month, day of week, hour of day, weekend indicator, season, and holiday flags.
- **Spatial features:** police district, neighborhood, and distance-based proximity to the nearest liquor store.
- **Crime-type features:** violent vs. property crime indicators and severity scores.

Temporal Crime Pattern Analysis (RQ1)

Research Question 1: *What are the temporal crime patterns in Detroit, and which months, days, and hours exhibit the highest crime rates?*

To address RQ1, we combined visual exploratory analysis with basic time-series statistics, following frameworks for crime trend and seasonality analysis [4, 2].

- We constructed multiple time series aggregated at daily, weekly, and monthly resolutions. Using `matplotlib`, `seaborn`, and `plotly`, we produced line plots for long-term trends.
- Calendar-style heatmaps revealed recurring weekly patterns. Cyclic behavior was examined with 24-hour plots and hour-by-day heatmaps.

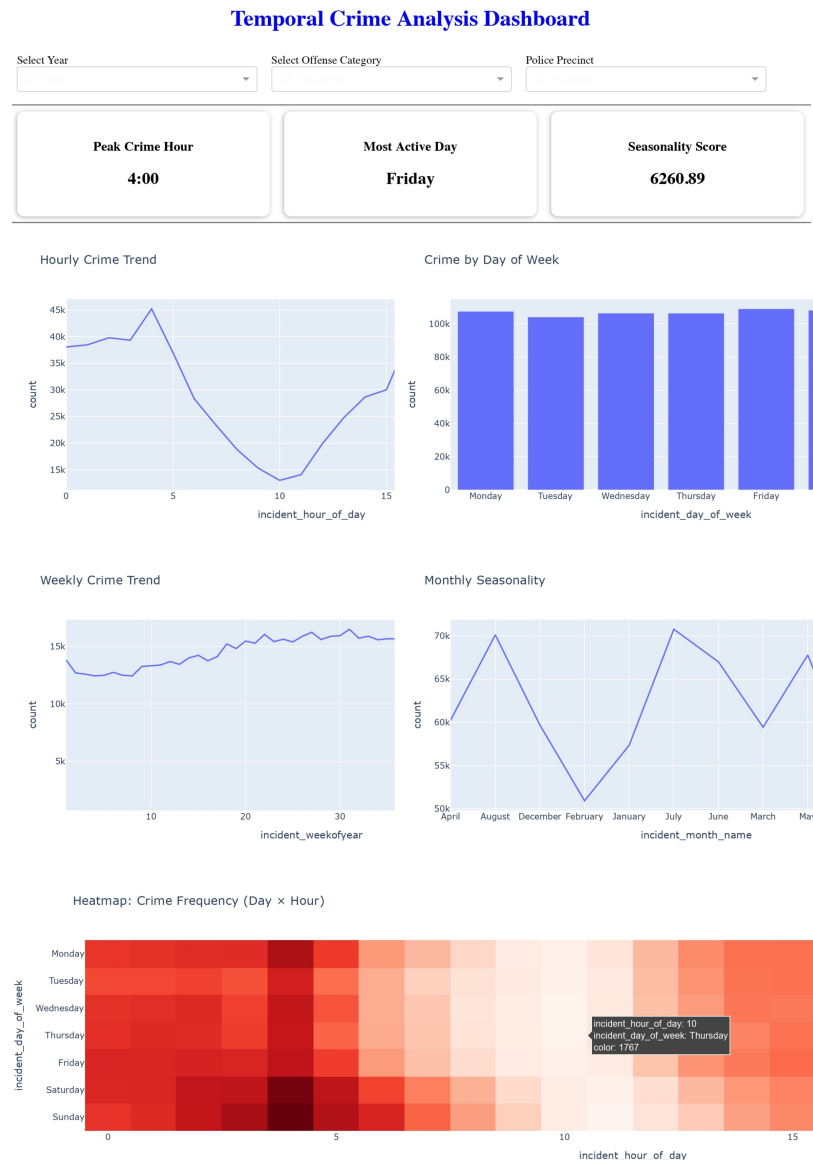


Figure 1: Interactive temporal crime analysis dashboard summarizing hourly, weekly, and monthly crime patterns in Detroit.

Spatial Crime Pattern Analysis (RQ2)

Research Question 2: *What police districts and geographic clusters are crime hotspots, and how does crime density vary across the city?*

For RQ2 we implemented a spatial analysis pipeline combining hotspot mapping, spatial statistics, and cluster detection, guided by Geographic Information System (GIS) based crime pattern studies in Stockholm, Detroit, and Austin [3, 6, 7, 5].

Hotspot Mapping and District Risk. Crime incidents were spatially joined to police precinct polygons using `geopandas`. We computed district-level crime rates and visualized them as choropleth maps overlaid with proportional-risk markers to distinguish low-, medium-, and high-risk precincts. Kernel Density Estimation (KDE) heatmaps and hexbin maps were generated to capture continuous spatial intensity patterns across Detroit.

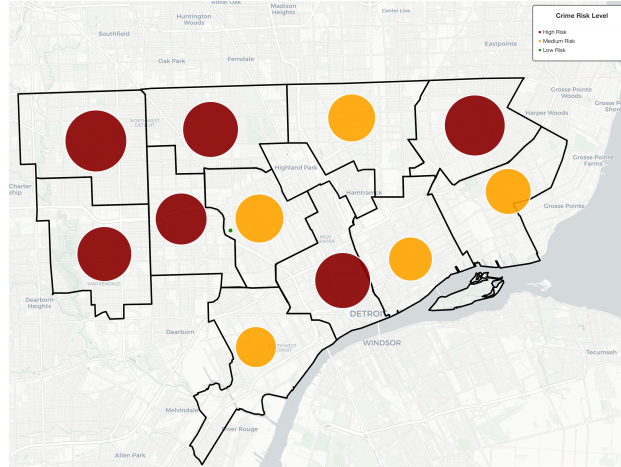
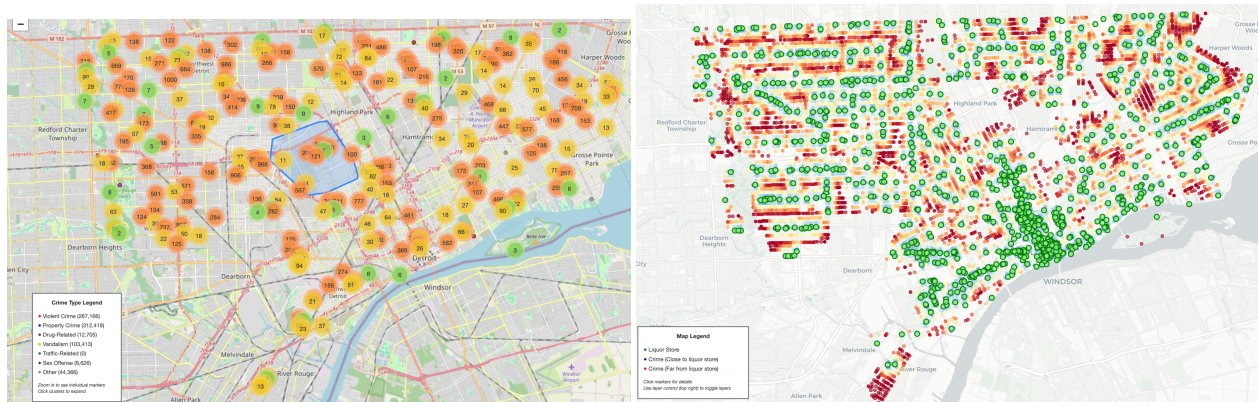


Figure 2: District-level crime risk visualization with high-, medium-, and low-risk precincts highlighted.

Point-Level Patterns and Clustering. Incident locations were plotted as interactive point maps with type-specific coloring and clustering enabled through **Leaflet/folium**. We applied Density-based spatial clustering of applications with noise (DBSCAN) to identify irregularly shaped clusters and filter out noise points, tuning the `eps` and `min_samples` parameters iteratively.

Liquor Stores and Crime Proximity. To explore the relationship between alcohol availability and crime, we computed distances from each crime to its nearest liquor store and visualized crimes as “near” vs. “far” from such establishments. This mapping, combined with density overlays of liquor outlets themselves, provides descriptive evidence consistent with prior findings that bars and alcohol-serving venues may co-locate with elevated crime risks [5].



(a) Incident clustering and hotspot identification using DBSCAN and interactive marker aggregation. (b) Overlay of crime incidents and liquor store locations, distinguishing crimes close to and far from liquor outlets.

Figure 3: Spatial crime visualizations supporting hotspot identification and crime-liquor proximity analysis.

Remaining Work Plans

Predictive Modeling of Crime Trends

Research Question 3 (RQ3): *How accurately can statistical and deep learning models predict future crime trends, and which forecasting method provides the most reliable predictions?*

Our next phase focuses on implementing and comparing multiple forecasting models, building on the time-series literature that combines statistical and deep learning methods for crime prediction [4, 8].

1) Baseline Statistical Models: We plan to implement one amongst three baseline models:

- **Seasonal Autoregressive Integrated Moving Average (SARIMA):** suitable for series with clear seasonality.
- **Prophet:** a decomposition-based model designed to handle missing values, multiple seasonalities, and holiday effects.
- **Holt–Winters (HWES):** a simpler exponential smoothing approach capturing level, trend, and seasonality.

2) Deep Learning Models: Utilize Long Short-Term Memory-based (LSTM) architectures in **TensorFlow** or **PyTorch**, including standard and Bidirectional LSTMs.

Resource Allocation and Predictive Policing Considerations

Research Question 4 (RQ4): *Can district-level crime forecasts enable targeted resource allocation, and which districts are projected to require increased police presence in the next 1–5 years?*

Building on the outputs from RQ3, RQ4 focuses on designing an analytical and visualization framework for data-driven resource allocation. Given the sensitive nature of predictive policing, this component comes with explicit attention to transparency, caution, and the avoidance of prescriptive interpretations.

Analytical Objectives. The planned analysis consists of four steps:

1. **Identify high-risk districts** using predicted crime trends and confidence intervals.
2. **Analyze trend direction** to determine which districts show stable, increasing, or decreasing trajectories.
3. **Generate resource allocation scenarios** rather than recommendations, including:
 - areas that *may require* increased patrol presence,
 - optimal timing windows based on temporal peaks,
 - seasonal considerations linked to historical patterns.

Ethical and Interpretive Considerations. Contemporary literature emphasizes the risks of algorithmic bias, feedback loops, and community impact in predictive policing systems[9]. We will explicitly address the following:

The purpose of RQ4 is not to prescribe policing strategies but to demonstrate how data-driven tools can be used responsibly and cautiously when exploring public safety resource questions. Ethical considerations will therefore be included in the final report to prevent misinterpretation of the analysis.

Timeline for Remaining Work:

- **Week 1:** Complete forecasting models (RQ3), compute district-level projections, and derive the risk index.
- **Week 2:** Develop interactive dashboard components, document ethical considerations, finalize scenario visualizations, and integrate all components into the final report and presentation.

References

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- [2] Timothy C.Hart. “Investigating Crime Pattern Stability at Micro-Temporal Intervals: Implications for Crime Analysis and Hotspot Policing Strategies”. In: *Criminal Justice Review* 46 (2021), pp. 173–189.
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